HyperRAG-IHM: Incremental Hyperedge Maintenance for Hypergraph RAG under Corpus Expansion

Haoyu Chen[1] **Jihua Zhu**[1,2*] **Tingkai Jia**[4] **Shengwei Zhao**[2,3] **Lin Chen**[1] **Xiaoxuan Mu**[1*]
[1]School of Software Engineering, Xi'an Jiaotong University
[2]State Key Laboratory of Human-Machine Hybrid Augmented Intelligence
[3]Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University
[4]East China Normal University
{haoyuchen01,zhaoshengwei,xiaoxuanmu}@stu.xjtu.edu.cn, zhujh@xjtu.edu.cn
51275902086@stu.ecnu.edu.cn, chenlin21@hnu.edu.cn

Abstract

Hypergraph-based methods extend structured retrieval-augmented generation (RAG) by organizing external knowledge as multi-entity relational structures, helping large language models (LLMs) generate better-grounded answers. However, existing structured RAG systems typically assume static corpus construction, requiring the retrieval structure to be rebuilt over the expanded corpus when new documents arrive. To address this limitation, we introduce **HyperRAG-IHM**, a hypergraph-based RAG framework that performs incremental hyperedge maintenance to incorporate newly arriving knowledge without reconstructing the historical retrieval structure. HyperRAG-IHM links relational evidence extracted from new documents to existing hypergraph knowledge, avoiding reconstruction of the entire retrieval structure. It uses LSH to localize candidate historical hyperedges, then applies multi-dimensional comparison and a three-action decision policy to integrate new relational evidence. Experiments show that HyperRAG-IHM achieves answer quality comparable to or better than full reconstruction under 5D scoring and 8D pairwise evaluation, and remains robust across ablation, structural behavior, cross-dataset, and cross-answer-model analyses.

1 Introduction

Retrieval-augmented generation (RAG) serves as a core paradigm for knowledge-intensive NLP tasks by enabling large language models (LLMs) to generate responses grounded in external knowledge rather than relying solely on parametric memory (Lewis et al., 2020; Guu et al., 2020; Izacard and Grave, 2021; Borgeaud et al., 2022; Gao et al., 2023). Such external grounding is especially important for reducing hallucination in knowledge-intensive generation (Ji et al., 2023). Early RAG systems relied on dense passage retrieval (Karpukhin et al., 2020). Recent work has moved toward structured retrieval methods, including graph-based, hierarchical, and memory-inspired approaches (Pan et al., 2024; Edge et al., 2024; Guo et al., 2025; Sarthi et al., 2024; Gutiérrez et al., 2024; Peng et al., 2025), which explicitly capture relational knowledge beyond flat text chunks. Hypergraph-based RAG further extends this direction by representing multi-entity associations as hyperedges (Feng et al., 2026, 2019).

In real-world deployments, domain knowledge does not remain fixed, and retrieval systems often need to operate over collections where new information arrives over time (Hou et al., 2025). Fresh clinical case reports, newly published biomedical literature, revised guidelines, and other domain-specific documents continuously change the evidence available to the system. However, these structured RAG systems are still largely built around static corpus construction (Zhang et al., 2025b). How to update the retrieval structure while preserving its relational organization remains underexplored when new documents arrive. A retrieval system that cannot absorb these updates gradually loses coverage, while incorporating new evidence without explicit maintenance can introduce duplicate or conflicting relational evidence into the retrieval structure. We refer to this requirement as **corpus expansion**: an existing knowledge base must absorb newly arrived documents over time while keeping its structured retrieval representation reliable.

For structured RAG systems, corpus expansion creates a tension between update cost and representation reliability. A recent preprint on temporal RAG notes that static-corpus assumptions leave update costs and retrieval stability insufficiently examined (Han et al., 2025). A recent preprint, EraRAG, builds a unified retrieval index over the expanded corpus, but its full-reconstruction setting

*Corresponding authors.

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, pages 20682–20693
October 24-29, 2026 ©2026 Association for Computational Linguistics

repeats extraction and construction over historical documents, causing update cost to grow with accumulated corpus size (Zhang et al., 2025b). Avoiding full reconstruction still requires deciding how newly extracted relational evidence should connect to the existing retrieval structure. Without explicit maintenance, these new-old relationships remain implicit, making it difficult to control relational redundancy and preserve coherent retrieved context as updates accumulate. The central challenge is therefore:

> *Can we build a RAG system whose knowledge structure can be incrementally updated as new knowledge arrives, while retaining strong organization and retrieval quality?*

We develop **HyperRAG-IHM**, an incremental hyperedge maintenance framework for hypergraph-based RAG. HyperRAG-IHM treats relational evidence extracted from newly arrived documents as maintenance targets and links it to existing hypergraph knowledge. It first uses an auxiliary LSH index to localize candidate historical hyperedges, and then compares new and existing relation records with entity, semantic, keyword, source, support, and specificity signals. The resulting maintenance decisions update the hyperedge layer through aggregation, replacement, or separate retention, allowing the retrieval structure to absorb new knowledge while controlling relational redundancy. We instantiate HyperRAG-IHM on HyperRAG, whose hyperedges provide concrete relation-level records containing entity sets, descriptions, keywords, source identifiers, and weights (Feng et al., 2026, 2019).

Our contributions are:

- We propose **HyperRAG-IHM**, an incremental maintenance framework that combines LSH-based candidate localization, multi-dimensional hyperedge comparison, and a three-action maintenance policy.
- We identify corpus expansion in hypergraph-based RAG as an incremental hyperedge maintenance problem, with the goal of incorporating newly arrived knowledge without reconstructing the existing retrieval structure.
- Experiments covering quality, efficiency, structural behavior, ablation, cross-dataset generalization, and cross-model robustness show that HyperRAG-IHM substantially reduces update cost while preserving competitive full-reconstruction answer quality.

2 Related Work

2.1 Structured Retrieval-Augmented Generation

Early RAG paired dense passage retrieval with generation (Karpukhin et al., 2020; Lewis et al., 2020; Guu et al., 2020; Izacard and Grave, 2021). Benchmarks such as KILT further consolidated knowledge-intensive language tasks around shared evidence sources (Petroni et al., 2021). Recent systems increasingly use corpus structure to support relational and multi-hop reasoning (Peng et al., 2025). GraphRAG builds entity relation graphs and community reports (Edge et al., 2024; Traag et al., 2019), LightRAG combines graph and vector retrieval (Guo et al., 2025), RAPTOR constructs hierarchical summary trees (Sarthi et al., 2024), and HippoRAG coordinates LLMs, knowledge graphs, and graph search (Gutiérrez et al., 2024). HyperRAG further uses hypergraphs to model multiway relations and thematic associations (Feng et al., 2026, 2019). These methods motivate structured retrieval, but they mainly organize retrieval over a fixed corpus rather than maintaining hypergraph structure during corpus expansion.

2.2 Dynamic and Continual RAG

Self-RAG and a CRAG preprint study inference-time retrieval control, critique, and correction during generation (Asai et al., 2024; Yan et al., 2024). Beyond retrieval-specific methods, recent work studies reasoning efficiency and reliability through selective chain-of-thought compression and adaptive refinement (Zhang et al., 2026c). Recent preprints further explore adaptive refinement and late-stage error mitigation (Zhang et al., 2026a, 2025a), as well as explicit or coarse-to-fine reasoning over multimodal and tabular inputs (Zhang et al., 2026b; Huang et al., 2026). These approaches optimize inference-time computation or intermediate reasoning representations without maintaining the underlying retrieval structure; when the corpus itself expands, index-level maintenance remains a separate problem. Continual neural information retrieval studies retrieval systems that adapt over time, including definitions,

datasets, and evaluation protocols for lifelong settings (Hou et al., 2025). A recent temporal-RAG preprint also highlights update-cost and retrieval-stability concerns under evolving knowledge (Han et al., 2025). The EraRAG preprint studies corpus expansion at the index-construction level and uses hyperplane-based LSH to localize insertions in graph-based retrieval indexes (Zhang et al., 2025b). However, none of these methods addresses maintenance of structured hyperedge representations under corpus expansion, where new evidence must be reconciled with existing hyperedges.

2.3 Knowledge Maintenance, Entity Resolution, and LSH

Knowledge graph research studies dynamic updates through entity alignment and temporal link prediction (Sun et al., 2020; Yan et al., 2021; Jin et al., 2020), but these methods usually operate on binary graph edges rather than hyperedges among multiple entities. Entity resolution and record linkage use candidate blocking followed by multi-feature comparison (Christophides et al., 2020; Papadakis et al., 2020), providing a useful analogy for restricting and comparing candidate hyperedges. LSH supports efficient approximate search, near duplicate detection, and streaming data processing (Indyk and Motwani, 1998; Manku et al., 2007; Leskovec et al., 2020); SimHash and MinHash provide widely used similarity estimators (Charikar, 2002; Broder, 1997). In this paper, these ideas serve as background for candidate localization; the concrete maintenance mechanism is defined in Section 4.1.

3 Preliminaries and Problem Formulation

3.1 Hypergraph Representation and Hyperedge Types

HyperRAG (Feng et al., 2026) organizes extracted knowledge as a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes entity vertices and $\mathcal{E}$ denotes hyperedges over these vertices. In the hypergraph-based representation used in this work, the LLM extraction process produces entities, low-order hyperedges, and high-order hyperedges in the same extraction pass. Each hyperedge is defined by an entity set and is associated with a textual description, keywords, source identifiers, and an edge weight, so that it can encode a relation or association among a group of entities.

Low-order hyperedges capture direct entity-level relational evidence: when a small number of entities are explicitly related in a text chunk, the system extracts a low-order hyperedge recording the relational description, keywords, and strength. High-order hyperedges instead organize broader thematic associations among multiple related entities identified during the same extraction pass. During incremental updates, high-order hyperedges follow the inherited key-based aggregation path: records with the same normalized entity set are consolidated, while records with different entity sets remain separate. HyperRAG-IHM applies its LSH-based three-action maintenance specifically to low-order hyperedges, where newly extracted relational evidence is most likely to overlap with or refine existing records.

3.2 Incremental Hyperedge Maintenance Problem

Let ΔD_t denote the batch of new texts arriving at update step t. Before this update, the system maintains an existing HyperRAG index built from the previously observed corpus. Its graph component is denoted as $\mathcal{H}_{t-1} = (\mathcal{V}_{t-1}, \mathcal{E}_{t-1})$, where $\mathcal{V}_{t-1}$ contains extracted entity vertices and $\mathcal{E}_{t-1}$ contains the stored hyperedges. The system also maintains an auxiliary LSH maintenance state $\mathcal{S}_{t-1}$ for active low-order hyperedges, including their normalized records and indexing metadata used for local candidate retrieval.

Given ΔD_t, the update objective is to construct an expanded hypergraph $\mathcal{H}_t$ and a synchronized auxiliary state $\mathcal{S}_t$ without rebuilding the historical hypergraph from scratch, such that:

1. $\mathcal{H}_t$ incorporates the knowledge extracted from the expanded corpus;
2. redundant, overlapping, or superseded low-order hyperedges are maintained through aggregation, replacement, or separate retention rather than being accumulated without control;
3. extraction and maintenance are restricted to ΔD_t and the hyperedges extracted from it, without re-running extraction over the historical corpus.

Each extracted low-order hyperedge is normalized as

$$e = (S_e, d_e, K_e, C_e, w_e), \tag{1}$$

where S_e is the entity set, d_e is the textual description, K_e is the keyword set, C_e is the set of source identifiers, and w_e is the edge weight. The canonical edge key is defined as the ordered tuple obtained by sorting the entity set:

$$\kappa(e) = \mathrm{sort}(S_e). \tag{2}$$

This key ensures that all records over the same entity set map to the same maintenance unit regardless of their original extraction order or source document.

Let $\Delta\mathcal{E}_t^{\mathrm{low}}$ denote the new low-order hyperedges extracted from ΔD_t. For each $e_{\mathrm{new}} \in \Delta\mathcal{E}_t^{\mathrm{low}}$, the goal is to select an action $a \in \{\textsc{Merge}, \textsc{Replace_old}, \textsc{Keep_both}\}$ with respect to existing hyperedges in the current index.

4 HyperRAG-IHM

For each newly extracted low-order hyperedge, HyperRAG-IHM localizes candidate historical edges, compares them with interpretable relational signals, and applies one of three maintenance actions. The resulting update is committed jointly to the main hypergraph and the auxiliary LSH state, so that subsequent updates operate on a synchronized structure. Figure 1 illustrates the overall workflow.

4.1 LSH-Based Candidate Retrieval

Motivated by EraRAG's use of LSH-based localization in corpus expansion (Zhang et al., 2025b), HyperRAG-IHM maintains an auxiliary LSH index that maps semantically similar edge representations to the same or nearby buckets (Indyk and Motwani, 1998), avoiding exhaustive comparison between each new edge and all historical edges. For each new low-order hyperedge, the index returns a bounded candidate set including matches under the canonical entity-set key $\kappa(e)$, nearby LSH buckets, and entity-overlap candidates; merge and replacement decisions are made by the comparison and decision modules below.

The index reuses a fixed random hyperplane matrix $P \in \mathbb{R}^{k\times d}$ across incremental stages and maintains bucket labels for active low-order hyperedges. This sidecar state provides stable locality preserving access to historical edges while leaving the main hypergraph as the authoritative retrieval structure.

Signature Computation. Following random-hyperplane similarity estimation (Charikar, 2002), each low-order hyperedge is projected into a binary signature space. For each hyperedge e, let x_e denote its textual representation:

$$x_e = [K_e \texttt{ [SEP] } \mathrm{sort}(S_e) \texttt{ [SEP] } d_e]. \tag{3}$$

The representation x_e is encoded by the embedding model into a vector $\mathbf{v}_e \in \mathbb{R}^d$, where d is the embedding dimension, and normalized as $\hat{\mathbf{v}}_e = \mathbf{v}_e/\|\mathbf{v}_e\|$. The k-bit signature is

$$\mathbf{b}(e) = \left[\mathbf{1}[\mathbf{p}_1^\top \hat{\mathbf{v}}_e > 0], \ldots, \mathbf{1}[\mathbf{p}_k^\top \hat{\mathbf{v}}_e > 0]\right] \tag{4}$$

where $\mathbf{p}_i$ is the i-th row of P and $\mathbf{1}[\cdot]$ is the indicator function. The binary vector is converted to a bucket label $\ell_e \in \{0, \ldots, 2^k-1\}$ used for indexing.

Candidate Retrieval. For a new edge e_{new}, candidates are retrieved from four sources in priority order: matches under $\kappa(e_{\mathrm{new}})$, same-bucket edges, Hamming-neighbor buckets within a fixed radius (Indyk and Motwani, 1998), and an entity-Jaccard fallback when bucket retrieval returns too few candidates (Broder, 1997). Duplicates are removed and the final candidate set is capped by a fixed maximum size. The retrieved candidates are then ranked by the prefilter score in Section 4.2.

4.2 Multi-Dimensional Edge Comparison

After candidate retrieval, each candidate pair $(e_{\mathrm{new}}, e_{\mathrm{old}})$ is represented by a set of interpretable comparison features. This design follows the broad entity-resolution pattern of blocking/filtering followed by multi-feature comparison (Christophides et al., 2020; Papadakis et al., 2020). Conceptually, these deterministic signals cover three aspects of a maintenance decision: identity and overlap, semantic compatibility, and evidence strength.

Entity, keyword, and source overlaps are computed with set-similarity measures such as Jaccard similarity. Containment and overlap-coefficient signals are included because two related hyperedges may not have identical entity sets but can still describe nested or partially overlapping relations (Broder, 1997). Semantic similarity uses the same textual edge representation as LSH signatures, allowing relation meaning to be compared beyond exact keyword or entity overlap. The resulting signals include entity overlap, overlap coefficient, keyword overlap, source overlap, source gain, semantic similarity, and containment. Support $q_{\mathrm{sup}}(e)$ is derived from source count and edge

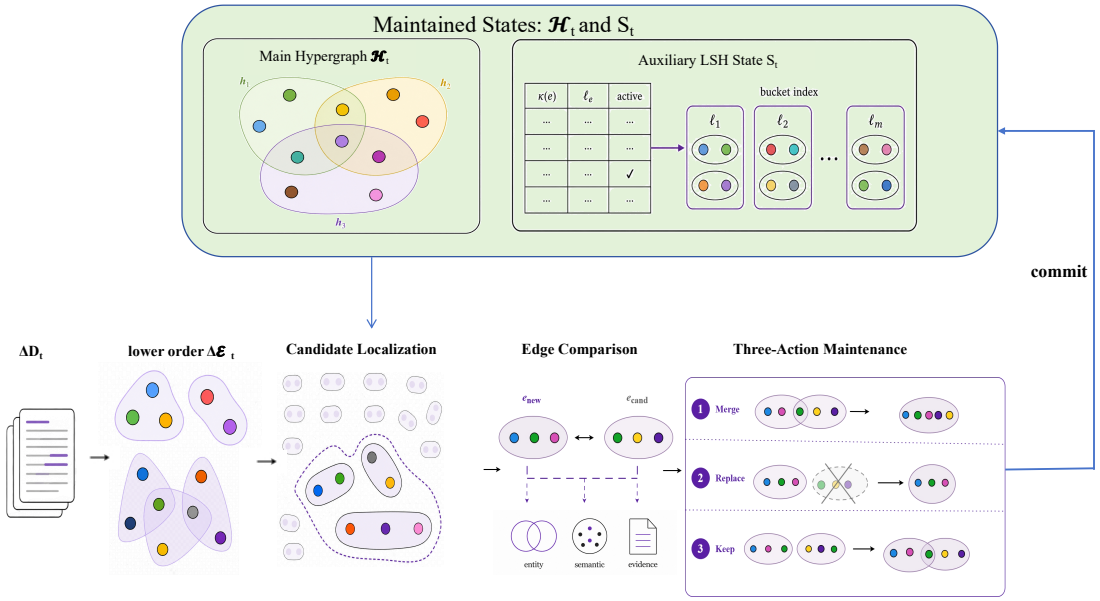


Figure 1: Overview of the HyperRAG-IHM framework.

weight, while specificity $q_{\text{spec}}(e)$ reflects informative descriptions and keywords. These features are combined into a prefilter score s_{pre} for candidate ranking, support and specificity deltas Δ_{sup} and Δ_{spec}, and composite merge and replacement scores s_{merge} and s_{replace} used by the decision mechanism below. Complete threshold values are listed in Appendix A.

4.3 Three-Action Decision Mechanism

Given the top-ranked candidate e^*_{old}, the system applies a deterministic decision tree based on the comparison features defined in Section 4.2. Here, S_{new} and S_{old} denote the entity sets of e_{new} and e^*_{old}, respectively. The three actions cover compatible, superseding, and distinct relations between new and existing hyperedges. The tree uses entity and semantic similarity, merge and replacement scores, and support/specificity deltas. If no candidate exists, the new edge is kept. When the top candidate shares the same sorted entity set, the system first checks replacement; if replacement is not triggered, the payloads are merged as repeated evidence for the same relation. When the top candidate has a contained or highly compatible entity set, the system can also merge the related evidence according to semantic and merge scores. If merge conditions are not met, the system checks replacement and then defaults to keeping both when similarity or evidence strength is insufficient.

Merge (aggregate compatible evidence). Merge aggregates compatible evidence into an existing maintenance unit. It applies to repeated evidence over the same sorted entity set and to contained or highly compatible candidates such as $S_{\text{new}} \subset S_{\text{old}}$. The decision uses semantic and merge scores together with supporting cues such as keyword overlap, source overlap, or no specificity increase. It deduplicates and concatenates descriptions, unions keyword and source sets, and sums weights.

Replace_old (new evidence supersedes old). Replacement is triggered when strong entity and semantic agreement is paired with a support or specificity advantage, or when reverse containment $S_{\text{old}} \subset S_{\text{new}}$ indicates a more complete and better supported relation. If the canonical keys differ, the old edge is marked inactive and the new edge is inserted; if they match, the maintenance unit is updated with the new description while carrying over compatible keyword/source evidence.

Keep_both (default). This default is applied when entity or semantic similarity is low, or when no candidate satisfies the merge or replacement conditions. It preserves potentially novel relations when aggregation or supersession is not sufficiently supported. Appendices A and B provide threshold values and concrete decision-log examples.

4.4 Consistency: Staged Update and Commit

All updates follow a staged commit protocol: decisions first produce upsert and removal lists, $\mathcal{U}$ and $\mathcal{R}$, from the current auxiliary state; the main hypergraph is then updated; and the auxiliary LSH state is committed only after successful write-back. If the main-graph write fails, the auxiliary state remains unchanged and the stage can be retried. This protocol prevents the auxiliary index from advancing beyond the persisted hypergraph state. After candidate retrieval, the pairwise scoring and decision cost per batch is $O(|\Delta\mathcal{E}_t^{\text{low}}| \cdot M)$, where M is the candidate cap. This does not remove extraction cost for newly arrived documents, but avoids detailed comparison against all historical low-order edges and avoids re-extracting the historical corpus.

5 Experiments and Analysis

Our experiments evaluate HyperRAG-IHM from four complementary perspectives. First, we test whether incremental maintenance preserves answer quality against full reconstruction. Second, we measure whether this quality is achieved with lower update cost. Third, we examine whether the proposed actions maintain the relationship layer rather than merely appending new edges. Finally, we evaluate whether the effect generalizes across datasets, answer models, and ablation settings.

5.1 Experimental Setup

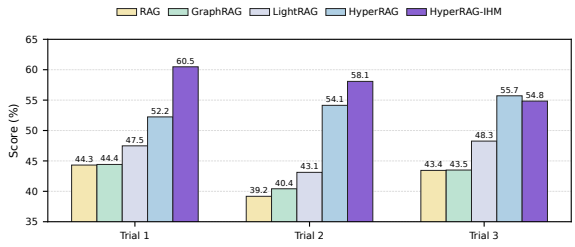
We summarize the shared experimental context in this subsection, while task-specific question splits and reporting details are introduced in the corresponding experiment subsections.

Datasets. We evaluate the proposed method on four datasets with different scale and domain characteristics. Pathology contains 5,297 medical pathology documents and 5,289 indexed chunks, and serves as the primary dataset for the detailed quality, robustness, maintenance, ablation, and efficiency analyses. Neurology contains 12,370 medical neurology documents and 12,370 indexed chunks, providing a larger medical-domain generalization setting. MixCorp contains 687 mixed-domain input documents and 817 indexed chunks derived from heterogeneous QA sources, providing an additional cross-domain setting. MuSiQue (Trivedi et al., 2022) is evaluated using a difficulty-stratified 40% subset containing 11,960 Wikipedia passages and 150 multi-hop questions, providing a larger open-domain generalization setting.

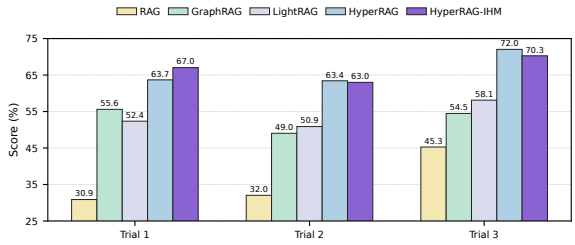
Compared systems. The main method is **HyperRAG-IHM** (*HyperRAG with Incremental Hyperedge Maintenance*), which updates an existing HyperRAG index by processing newly arrived texts and their extracted hyperedges. The primary reference is **HyperRAG** full reconstruction, which rebuilds the HyperRAG index from the expanded corpus and is used to assess whether incremental maintenance preserves full-reconstruction quality. For broader context on the primary dataset, we also include **GraphRAG** (Edge et al., 2024), **LightRAG** (Guo et al., 2025), and an embedding-only chunk-retrieval RAG baseline (Lewis et al., 2020; Karpukhin et al., 2020). Their results serve as reference points on the primary dataset.

Evaluation protocol. We adopt the evaluation protocol used by HyperRAG (Feng et al., 2026). Specifically, its **5D scoring-based evaluation** extends the GraphRAG evaluation paradigm based on LLM judges (Edge et al., 2024; Zheng et al., 2023; Liu et al., 2023) to measure absolute answer quality across *Comprehensiveness*, *Diversity*, *Empowerment*, *Logical*, and *Readability*. Its **selection-based evaluation** compares two answers across eight dimensions—*Comprehensiveness*, *Empowerment*, *Accuracy*, *Relevance*, *Coherence*, *Clarity*, *Logical*, and *Flexibility*—and records which answer is judged better in each dimension. In this work, 5D scoring is used for broad absolute-quality comparisons, whereas 8D pairwise selection is used for fine-grained relative comparison: by comparing two answers under the same judge prompt, it better exposes relative answer-quality differences and reduces perturbations from score-scale calibration and answer style. For 8D selection, we report clear-preference rates.

Model settings. All HyperRAG index construction and hyperedge extraction use Doubao-1.5-Lite as the construction LLM. Doubao-1.5-Lite serves as the primary judge for the reported evaluations. To assess judge sensitivity, we additionally evaluate the Pathology comparison with DeepSeek-V3.2, GLM-5, and Kimi-K2.5. For answer generation, the main experiments use Doubao-1.5-Lite, and cross-answer-model analyses additionally use MiniMax-M2.7, Kimi-2.5,



(a) Single-hop setting.



(b) Multi-hop setting.

Figure 2: 5D average scores on Pathology.

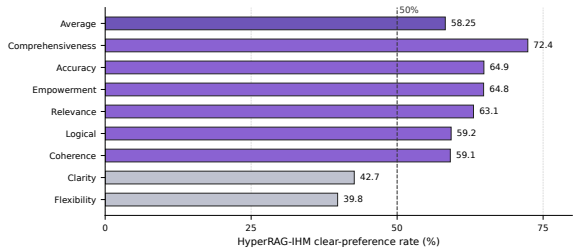


Figure 3: 8D pairwise preference between HyperRAG-IHM and HyperRAG.

and Qwen3.6-Plus.

5.2 Primary-Dataset Overall Comparison

Pathology serves as the primary dataset, with dense medical entities, specialized terminology, and clinically grounded relations that require factual grounding and evidence integration. We evaluate single-hop questions for ordinary factual retrieval and multi-hop questions requiring joint evidence across texts. Across three repeated trials, we sample 100 single-hop and 50 multi-hop questions per trial; Figures 2a and 2b report the 5D averages, with detailed scores in Appendix C.

HyperRAG-IHM achieves the highest 5D average score in both single-hop (57.80) and multi-hop (66.77) settings, with gains mainly in Comprehensiveness, Logical, and Readability. We further compare it with the HyperRAG full-reconstruction reference using the full 8D pairwise preference evaluation.

Figure 3 shows that HyperRAG-IHM is preferred in 58.25% of clear-preference comparisons, with the strongest advantages in Comprehensiveness (72.4%), Accuracy (64.9%), and Empowerment (64.8%). These results indicate that HyperRAG-IHM preserves competitive full-reconstruction quality while showing advantages in several coverage-oriented dimensions under the primary judge.

5.3 Efficiency Analysis

We evaluate efficiency using a single 5% corpus update on Pathology as an example. Non-incremental baselines rebuild their graph or index over the expanded corpus, whereas HyperRAG-IHM processes newly arrived documents and maintains their extracted hyperedges. As shown in Table 1, HyperRAG-IHM requires roughly 20× fewer LLM tokens and 19× less wall-clock time than HyperRAG full reconstruction, indicating that incremental maintenance substantially reduces update cost while preserving quality.

Method	Tokens (M)	Time (s)
HyperRAG-IHM	1.63	1,221.06
LightRAG	32.51	1,554.29
GraphRAG	47.46	10,334.08
HyperRAG	32.95	22,947.62

Table 1: Construction cost for a 5% Pathology corpus update. Non-incremental methods rebuild their graph or index.

5.4 Maintenance Behavior and Structure Size

To connect answer quality with structural behavior, we examine the actions taken during incremental maintenance and the resulting low-order relationship-layer size. Table 2 reports merge and replacement counts, together with the final maintained relationship entries.

Dataset	Merge	Repl.	IHM Rel.	H-RAG Rel.	Red.
Pathology	423	128	25,353	32,029	20.8%
Neurology	1,091	360	75,227	90,940	17.3%

Table 2: Maintenance actions and low-order relationship-layer size. Repl.: replacements; IHM Rel. and H-RAG Rel.: retained low-order relationship entries for HyperRAG-IHM and HyperRAG full reconstruction; Red.: reduction of IHM relative to HyperRAG.

On both datasets, merge actions substantially

outnumber replacements, and the maintained relationship layer is 17–21% smaller than the full-reconstruction reference. This indicates that incremental maintenance consolidates redundant relational evidence while preserving distinct relationships through the keep-both fallback.

5.5 Ablation Study

To isolate the proposed maintenance mechanism, we compare HyperRAG-IHM with two increasingly stronger incremental baselines. *Direct Insert* adds new hyperedges without maintenance. *Key-based Aggregation* is a stronger ablation that aggregates incoming hyperedges when they share normalized entity keys. HyperRAG-IHM further introduces LSH-based candidate retrieval, multi-dimensional comparison, and explicit maintenance actions.

The complete 5D ablation scores are provided in Appendix D. HyperRAG-IHM obtains the highest 5D average in both settings, with a larger improvement on multi-hop questions. This suggests that active hyperedge maintenance is especially useful when answering requires combining relational evidence across documents. Table 3 reports the corresponding 8D pairwise preference rates against the two incremental baselines.

Baseline	Type	Avg.	Comp.	Acc.	Rel.	Logic.
Direct Insert	Single	57.51	73.49	66.67	59.93	58.51
Direct Insert	Multi	65.84	75.33	73.72	76.30	69.29
Key Aggreg.	Single	56.42	73.15	65.12	59.09	56.32
Key Aggreg.	Multi	65.07	75.33	70.50	74.81	66.90

Table 3: 8D pairwise ablation results. Values are HyperRAG-IHM preference rates among clear-preference cases. Comp., Acc., Rel., and Logic. denote Comprehensiveness, Accuracy, Relevance, and Logical.

The same trend appears in pairwise evaluation: HyperRAG-IHM is preferred over both Direct Insert and Key-based Aggregation, with stronger gains on multi-hop questions. The gap over Direct Insert shows the cost of leaving newly extracted relations unmaintained. The gap over Key-based Aggregation further shows that exact-key aggregation is insufficient; effective maintenance also requires candidate localization and semantic comparison across related but non-identical hyperedges.

5.6 Generalization and Robustness

We further evaluate whether the maintained hypergraph remains effective across datasets, answer-generation models, and judge models. Table 4 reports the cross-dataset results. On Neurology, HyperRAG-IHM obtains a 5D average score of 54.32, slightly above HyperRAG at 53.72, and reaches a 52.50% 8D clear-preference rate. On MixCorp, HyperRAG has a higher 5D average score (54.65 versus 53.39), while HyperRAG-IHM still achieves a 51.88% 8D clear-preference rate. For MuSiQue, we take 20% of the subset as the initial index and apply sixteen consecutive 5% updates before comparing the maintained index with full reconstruction. HyperRAG-IHM obtains a 5D average of 62.71 versus 61.26 for HyperRAG and a 51.10% 8D clear-preference rate; full dimension-level results are provided in Appendix E. Across the sixteen updates, active low-order hyperedges remain between 4.6 and 4.9 per chunk, with 1.45% variation, while the largest LSH bucket stays below 0.4% of indexed relationships.

Dataset	5D IHM	5D H-RAG	8D Avg.
Pathology	60.79	58.15	58.25
Neurology	54.32	53.72	52.50
MixCorp	53.39	54.65	51.88
MuSiQue	62.71	61.26	51.10

Table 4: Cross-dataset generalization. 8D values are HyperRAG-IHM clear-preference rates.

When only the answer-generation model changes, HyperRAG-IHM is preferred over HyperRAG with average 8D clear-preference rates from 62.72% to 65.72% across Kimi-2.5, Qwen3.6-Plus, and MiniMax-M2.7 (Table 5). Under these alternative answer models, the 8D pairwise advantage is even more pronounced than with the main model, suggesting that the maintained hypergraph remains useful beyond the main answer-model setting despite slightly lower absolute 5D scores.

Answer Model	5D IHM	5D H-RAG	8D Avg.
Kimi-2.5	73.94	75.46	65.72
Qwen3.6-Plus	66.34	67.89	62.72
MiniMax-M2.7	73.24	75.77	64.21

Table 5: Cross-answer-model robustness on Pathology. 8D values are HyperRAG-IHM clear-preference rates.

To assess sensitivity to the LLM judge, we re-evaluate the same 100 single-hop and 50 multi-hop Pathology answers with the primary judge and three independent judges. Table 6 shows that all four judges favor HyperRAG-IHM in 8D pairwise comparison, with clear-preference rates from 53.13% to 60.72%. The 5D ordering is more judge-sensitive: it favors HyperRAG-IHM under the primary judge but marginally favors HyperRAG under the three independent judges.

Judge	5D IHM	5D H-RAG	8D Avg.
Doubao-1.5-Lite	63.91	61.61	60.72
DeepSeek-V3.2	74.30	76.60	58.79
GLM-5	74.99	76.22	56.35
Kimi-K2.5	68.86	69.11	53.13

Table 6: Cross-judge robustness on the same Pathology subset. 8D values are HyperRAG-IHM clear-preference rates.

Taken together, the cross-dataset, cross-answer-model, and cross-judge results show competitive behavior across the evaluated settings, with a consistent pairwise advantage for HyperRAG-IHM.

6 Conclusion

We have proposed the first incremental low-order hyperedge maintenance method for hypergraph-based RAG, enabling knowledge bases to absorb new documents without full-graph reconstruction. The core mechanism combines LSH candidate retrieval, multi-dimensional scoring, and a three-action decision policy, preserving answer quality competitive with full reconstruction under LLM-judge evaluation. Cross-dataset, cross-answer-model, and cross-judge evaluations support the robustness of these findings, and the efficiency analysis confirms that incremental maintenance substantially reduces construction cost. Beyond HyperRAG, the maintenance framework could in principle be adapted to other structured RAG systems that maintain entity-level relational records, providing a foundation for scalable knowledge maintenance under corpus growth. Future directions include learning decision thresholds from data rather than hand-tuning them, extending three-action semantic maintenance to high-order thematic hyperedges, and evaluating on corpora with document revisions and temporal conflicts.

Limitations

The decision thresholds are empirically specified rather than optimized; the results demonstrate effectiveness under the present settings but do not prove optimality. Although the experiments simulate corpus expansion through incremental updates, they use static datasets; true dynamic corpora with document revisions, temporal conflicts, and freshness-sensitive queries remain beyond the current evaluation. Cross-judge evaluation reduces dependence on a single automatic evaluator, but the current study does not include human annotation.

Acknowledgments

This work was supported by the Natural Science Basis Research Plan in Shaanxi Province of China under Grant 2026JC-QYCX-061.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*.

Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego De Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, and 9 others. 2022. Improving language models by retrieving from trillions of tokens. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 2206–2240. PMLR.

Andrei Z. Broder. 1997. On the resemblance and containment of documents. In *Proceedings of the Compression and Complexity of Sequences 1997*, SEQUENCES '97, pages 21–29. IEEE Computer Society.

Moses S. Charikar. 2002. Similarity estimation techniques from rounding algorithms. In *Proceedings of the Thirty-Fourth Annual ACM Symposium on Theory of Computing*, STOC '02, pages 380–388, New York, NY, USA. Association for Computing Machinery.

Vassilis Christophides, Vasilis Efthymiou, Themis Palpanas, George Papadakis, and Kostas Stefanidis. 2020. An overview of end-to-end entity resolution for big data. *ACM Computing Surveys*, 53(6):1–42.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Yifan Feng, Hao Hu, Shihui Ying, Xingliang Hou, Shiquan Liu, Mingyuan Yang, Junchang Li, Shaoyi Du, Nanning Zheng, Han Hu, and Yue Gao. 2026. Hyper-RAG: Combating LLM hallucinations using hypergraph-driven retrieval-augmented generation. *Nature Communications*, 17:5778.

Yifan Feng, Haoxuan You, Zizhao Zhang, Rongrong Ji, and Yue Gao. 2019. Hypergraph neural networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 3558–3565.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2023. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2025. LightRAG: Simple and fast retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10746–10761, Suzhou, China. Association for Computational Linguistics.

Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 59532–59569.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. Retrieval augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*, pages 3929–3938.

Jiale Han, Austin Cheung, Yubai Wei, Zheng Yu, Xusheng Wang, Bing Zhu, and Yi Yang. 2025. RAG meets temporal graphs: Time-sensitive modeling and retrieval for evolving knowledge. *arXiv preprint arXiv:2510.13590*.

Jingrui Hou, Georgina Cosma, and Axel Finke. 2025. Advancing continual lifelong learning in neural information retrieval: Definition, dataset, framework, and empirical evaluation. *Information Sciences*, 687:121368.

Qixian Huang, Hongqiang Lin, Tong Fu, Yingsen Wang, Zhenghui Fu, Qirui Wang, Yiding Sun, and Dongxu Zhang. 2026. CFMS: A coarse-to-fine multimodal synthesis framework for enhanced tabular reasoning. *arXiv preprint arXiv:2604.10973*.

Piotr Indyk and Rajeev Motwani. 1998. Approximate nearest neighbors: Towards removing the curse of dimensionality. In *Proceedings of the Thirtieth Annual ACM Symposium on Theory of Computing*, STOC '98, pages 604–613, New York, NY, USA. Association for Computing Machinery.

Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 874–880, Online. Association for Computational Linguistics.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Woojeong Jin, Meng Qu, Xisen Jin, and Xiang Ren. 2020. Recurrent event network: Autoregressive structure inference over temporal knowledge graphs. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6669–6683, Online. Association for Computational Linguistics.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.

Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman. 2020. *Mining of Massive Datasets*, 3rd edition. Cambridge University Press, Cambridge, United Kingdom.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522.

Gurmeet Singh Manku, Arvind Jain, and Anish Das Sarma. 2007. Detecting near-duplicates for web crawling. In *Proceedings of the 16th International Conference on World Wide Web*, WWW '07, pages 141–150, New York, NY, USA. Association for Computing Machinery.

Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Jiapu Wang, and Xindong Wu. 2024. Unifying large language models and knowledge graphs: A roadmap.

IEEE Transactions on Knowledge and Data Engineering, 36(7):3580–3599.

George Papadakis, Dimitrios Skoutas, Emmanouil Thanos, and Themis Palpanas. 2020. Blocking and filtering techniques for entity resolution: A survey. *ACM Computing Surveys*, 53(2):1–42.

Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. 2025. Graph retrieval-augmented generation: A survey. *ACM Transactions on Information Systems*, 44(2):1–52.

Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, Majid Yazdani, Nicola De Cao, James Thorne, Yacine Jernite, Vladimir Karpukhin, Jean Maillard, Vassilis Plachouras, Tim Rocktäschel, and Sebastian Riedel. 2021. KILT: A benchmark for knowledge intensive language tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2523–2544.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. 2024. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *The Twelfth International Conference on Learning Representations*.

Zequn Sun, Chengming Wang, Wei Hu, Muhao Chen, Jian Dai, Wei Zhang, and Yuzhong Qu. 2020. Knowledge graph alignment network with gated multi-hop neighborhood aggregation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 222–229.

Vincent A. Traag, Ludo Waltman, and Nees Jan van Eck. 2019. From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9(1):5233.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*.

Yuchen Yan, Lihui Liu, Yikun Ban, Baoyu Jing, and Hanghang Tong. 2021. Dynamic knowledge graph alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 4564–4572.

Dongxu Zhang, Hongqiang Lin, Yiding Sun, Pengyu Wang, Qirui Wang, Ning Yang, and Jihua Zhu. 2026a. Not all queries need deep thought: CoFiCot for adaptive coarse-to-fine stateful refinement. *arXiv preprint arXiv:2603.08251*.

Dongxu Zhang, Yiding Sun, Pengcheng Li, Yumou Liu, Hongqiang Lin, Haoran Xu, Xiaoxuan Mu, Liang Lin, Wenbiao Yan, Ning Yang, Chaowei Fang, Juanjuan Zhao, Jihua Zhu, Conghui He, and Cheng Tan. 2026b. PointCoT: A multi-modal benchmark for explicit 3d geometric reasoning. *arXiv preprint arXiv:2602.23945*.

Dongxu Zhang, Yiding Sun, Cheng Tan, Wenbiao Yan, Ning Yang, Jihua Zhu, and Haijun Zhang. 2026c. Chain-of-thought compression should not be blind: V-skip for efficient multimodal reasoning via dual-path anchoring. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 19402–19415, San Diego, California, United States. Association for Computational Linguistics.

Dongxu Zhang, Yujun Wu, Yiding Sun, Jinnan Yang, Ning Yang, Jihua Zhu, Miao Xin, and Baoliang Tian. 2025a. Not all errors are created equal: ASCoT addresses late-stage fragility in efficient LLM reasoning. *arXiv preprint arXiv:2508.05282*.

Fangyuan Zhang, Zhengjun Huang, Yingli Zhou, Qintian Guo, Zhixun Li, Wensheng Luo, Di Jiang, Yixiang Fang, and Xiaofang Zhou. 2025b. EraRAG: Efficient and incremental retrieval augmented generation for growing corpora. *arXiv preprint arXiv:2506.20963*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623.

A Complete Threshold Parameters

Parameter	Value	Role
$\tau_{\text{ent}}^{\text{cand}}$	0.5	Entity Jaccard fallback
$\tau_{\text{sem}}^{\text{merge}}$	0.82	Semantic sim. for merge
$\tau_{\text{kw}}^{\text{merge}}$	0.4	Keyword overlap for merge
$\tau_{\text{sem}}^{\text{subset}}$	0.84	Sem. sim. for subset merge
$\tau_{\text{ent}}^{\text{replace}}$	0.8	Entity overlap for replace
$\tau_{\text{sem}}^{\text{replace}}$	0.75	Semantic sim. for replace
$\tau_{\text{sem}}^{\text{keep}}$	0.65	Below $\rightarrow$ Keep_both
$\tau_{\text{ent}}^{\text{keep}}$	0.5	Below $\rightarrow$ Keep_both
$\tau_{\text{merge}}^{\text{contain}}$	0.6	Containment merge score
$\tau_{\text{replace}}^{\text{contain}}$	0.4	Containment replace score
M_{extra}	1	Max extra entities for containment
M_{cand}	8	Candidate edge cap
k	16	LSH hyperplanes
r_{hamming}	2	Hamming search radius

Table 7: Complete threshold parameters.

B Decision Log Examples

Merge. An incoming edge about neurons and synapses matched an existing edge with the same entity key. The semantic and keyword scores were 0.91 and 0.62, respectively. The exact key and threshold conditions were both met, so descriptions were concatenated, source sets were merged, and weights were summed.

Replace. An incoming GABA inhibition edge matched an existing edge with entity and semantic scores of 0.85 and 0.79. The incoming edge had stronger support and specificity scores than the existing edge. The existing edge was marked inactive, and the incoming edge was inserted.

Keep. An incoming dopamine reward edge had dopamine movement as its top candidate, but the entity score was only 0.33. The low overlap triggered the conservative keep both action, preserving the two distinct relationships.

C Pathology 5D Detailed Scores

Method	Type	Comp.	Div.	Emp.	Logical	Read.	Avg.
RAG	Single	62.70	1.53	0.53	69.87	76.94	42.31
GraphRAG	Single	64.03	18.77	5.53	59.83	65.74	42.78
LightRAG	Single	60.63	27.17	15.97	57.17	70.50	46.29
HyperRAG	Single	76.14	31.04	11.15	75.32	76.52	54.03
HyperRAG-IHM	Single	**80.47**	**33.82**	13.97	**79.63**	**81.12**	**57.80**
RAG	Multi	56.00	3.00	1.13	58.80	61.41	36.07
GraphRAG	Multi	78.87	27.20	4.00	79.53	75.60	53.04
LightRAG	Multi	76.27	22.13	**14.13**	73.60	82.80	53.79
HyperRAG	Multi	94.13	**42.60**	12.07	93.25	89.87	66.38
HyperRAG-IHM	Multi	**95.73**	39.80	9.53	**95.60**	**93.21**	**66.77**

Table 8: Detailed 5D scores on Pathology. Single-hop scores are averaged over three 100-question trials, and multi-hop scores are averaged over three 50-question trials. Comp., Div., Emp., Read., and Avg. denote Comprehensiveness, Diversity, Empowerment, Readability, and Average.

D Ablation 5D Details

Method	Type	Comp.	Div.	Emp.	Logical	Read.	Avg.
Direct Insert	Single	77.60	**35.07**	11.90	75.98	77.27	55.56
Key-based Aggregation	Single	**80.47**	31.60	10.43	79.23	78.65	56.08
HyperRAG-IHM	Single	**80.47**	33.82	**13.97**	**79.63**	**81.12**	**57.80**
Direct Insert	Multi	89.67	**39.87**	7.93	89.87	89.17	63.30
Key-based Aggregation	Multi	90.93	39.53	**13.60**	91.03	87.23	64.47
HyperRAG-IHM	Multi	**95.73**	39.80	9.53	**95.60**	**93.21**	**66.77**

Table 9: 5D ablation results under single-hop and multi-hop settings.

E MuSiQue Pairwise Details

Method	Comp.	Emp.	Acc.	Rel.	Coh.	Clar.	Logic.	Flex.	Avg.
HyperRAG-IHM	71.33	66.00	57.25	52.45	48.95	30.20	50.00	32.62	51.10
HyperRAG	28.67	34.00	42.75	47.55	51.05	69.80	50.00	67.38	48.90

Table 10: Full 8D pairwise results on MuSiQue. Values are preference rates among clear-preference cases.